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Exercise 1 : SQL fundamentals- SELECT & Filtering

01. Selecting and Sorting

1. * General Syntax

SELECT col 1,

col 2,

col 3,

col 4,

col 5,

col 6,

col 7

FROM employees ;

* Results

Id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. * General Syntax

SELECT DISTINCT (department)

FROM employees ;

Results

department

IT

HR

Finance

Marketing

3. * General Syntax

SELECT first-name,
last-name

FROM employees

ORDER BY Salary DESC;

*Results

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. General Syntax

SELECT id,

first-name,

last-name,

Salary

FROM employees

ORDER BY Salary DESC

Limit 3 ;

*Results

id	first-name	last-name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

02 Filtering with WHERE

5. * General Syntax

```
SELECT id,
       first-name,
       last-name,
       department
```

FROM employees

WHERE department = 'IT';

Results

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. * General Syntax

```
SELECT id,
       first-name,
       last-name,
       department,
       Salary
```

FROM employees

WHERE department = 'finance' AND
Salary > 60000;

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65 000

7. *General Syntax

```
SELECT id,
       first-name,
       last-name,
       department
FROM employees
WHERE department = 'HR' OR
       department = 'Marketing';
```

Result

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. General Syntax

```
SELECT id,
       first-name,
       last-name,
       department
FROM employees
WHERE NOT department = 'IT';
```

Results

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. * General Syntax

```
SELECT id,  
       first-name,  
       last-name,  
       department
```

```
FROM employees
```

```
WHERE department IN ['IT', 'HR', 'Finance'] ;
```

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. * General Syntax

```
SELECT id,  
       first-name,  
       lastname,  
       department,  
       Salary,  
       City
```

```
FROM employees
```

```
WHERE department = 'IT'
```

```
Salary > 65000 AND
```

```
City = 'Johannesburg' ;
```

* Result

id	first-name	last-name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
5	Eva	Naidoo	IT	72 000	Johannesburg